

Determination of mechanical and physical properties of pig ear cartilagem through modeling and simulation using DFT

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1. INTRODUCTION

➤ **Pig ear cartilage** is commonly used as a biological model due to its similarity to human connective tissue. Its main structural protein, **elastin**, is responsible for tissue elasticity and resilience.

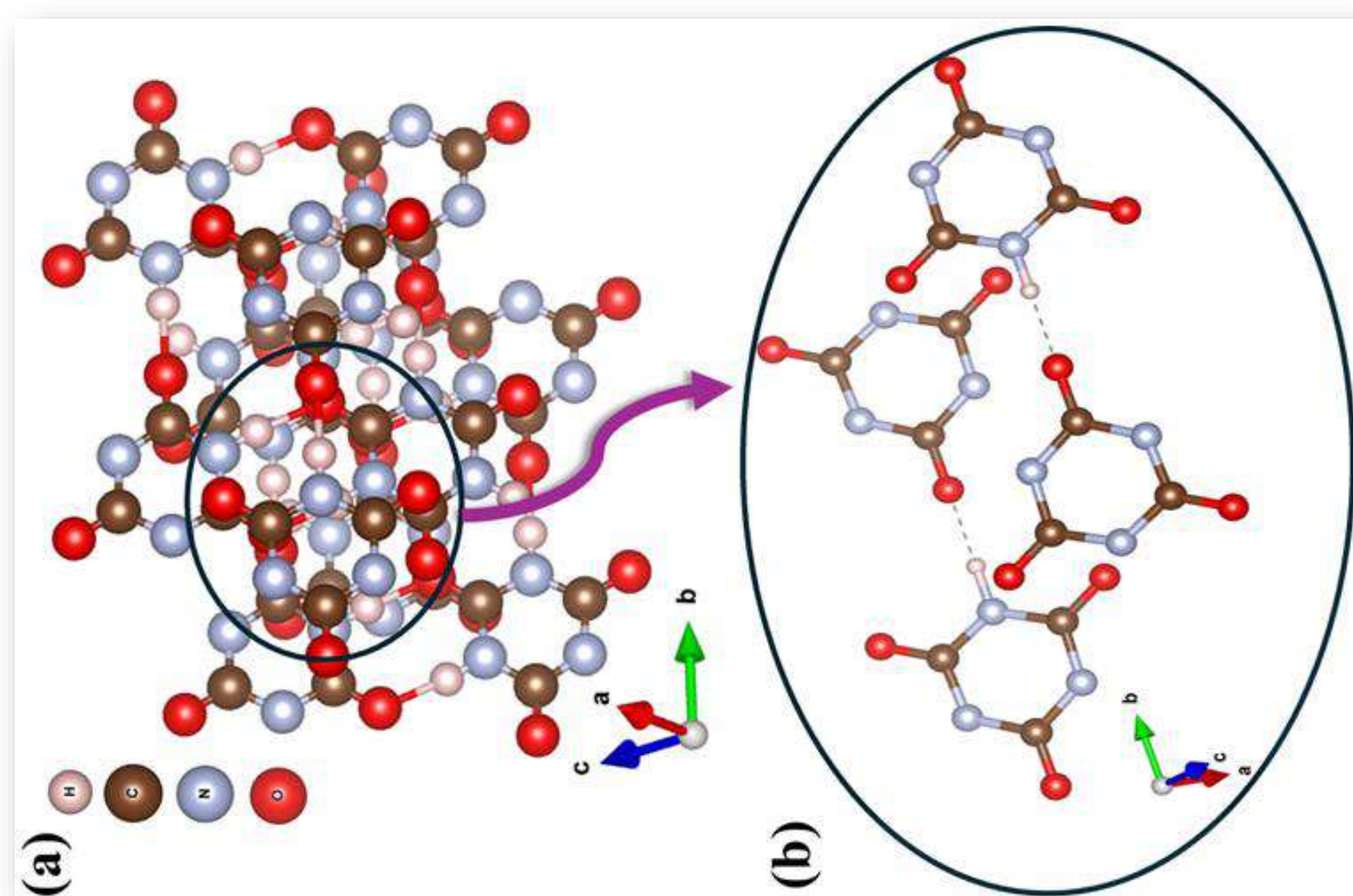


Figure 1: Optimized elastin fragment used in DFT calculations.

2. MATERIALS AND METHODS

➤ A **peptide-like elastin** molecular model, composed of **48 atoms** per unit cell, was constructed and optimized using first-principles DFT calculations. The structure crystallized in the monoclinic space group **C2/c**.

➤ **Structural optimization** → **Functional PBEsol**

➤ **Electronic properties:** total and partial densities of states (TDOS and PDOS).

➤ **Optical behavior:** dielectric function, absorption spectrum, refractive index, reflectivity and electron energy loss spectroscopy (EELS).

➤ **structural; electronic, optical properties**

3. RESULTS

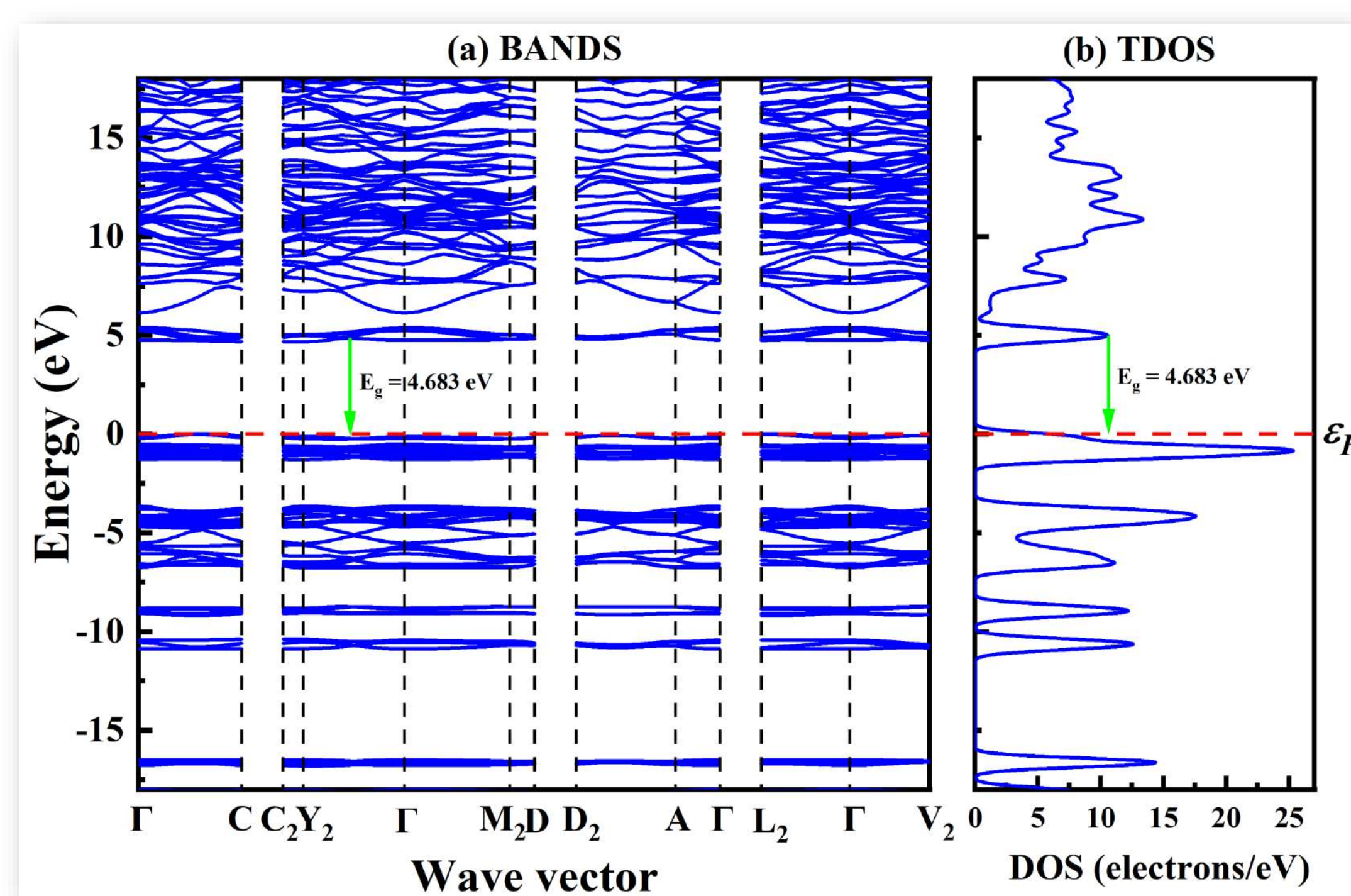


Figure 1: BANDS and TDOS

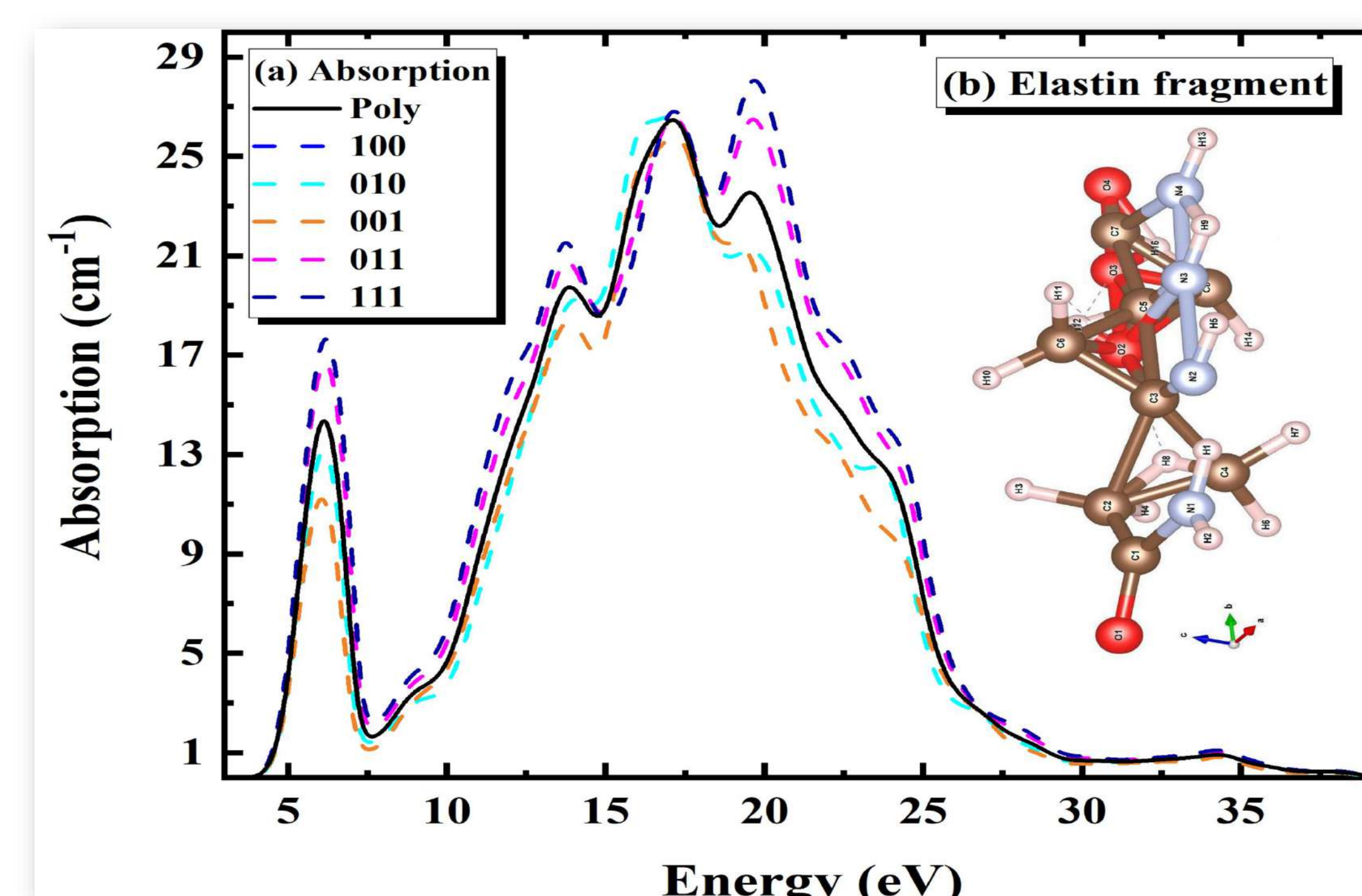


Figura 2. Optical Absorption

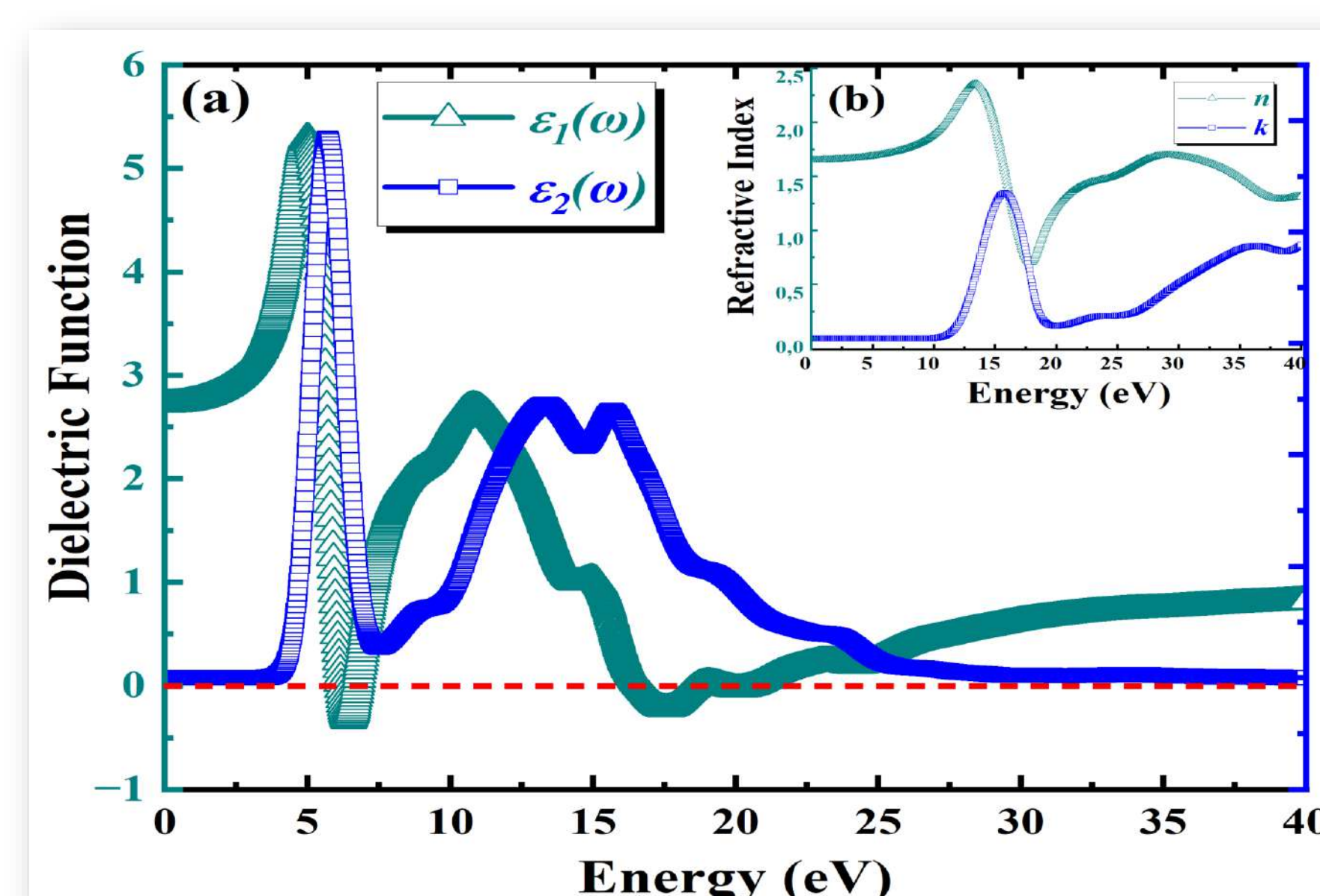


Figure 3. Dielectric Function and Refractive Index

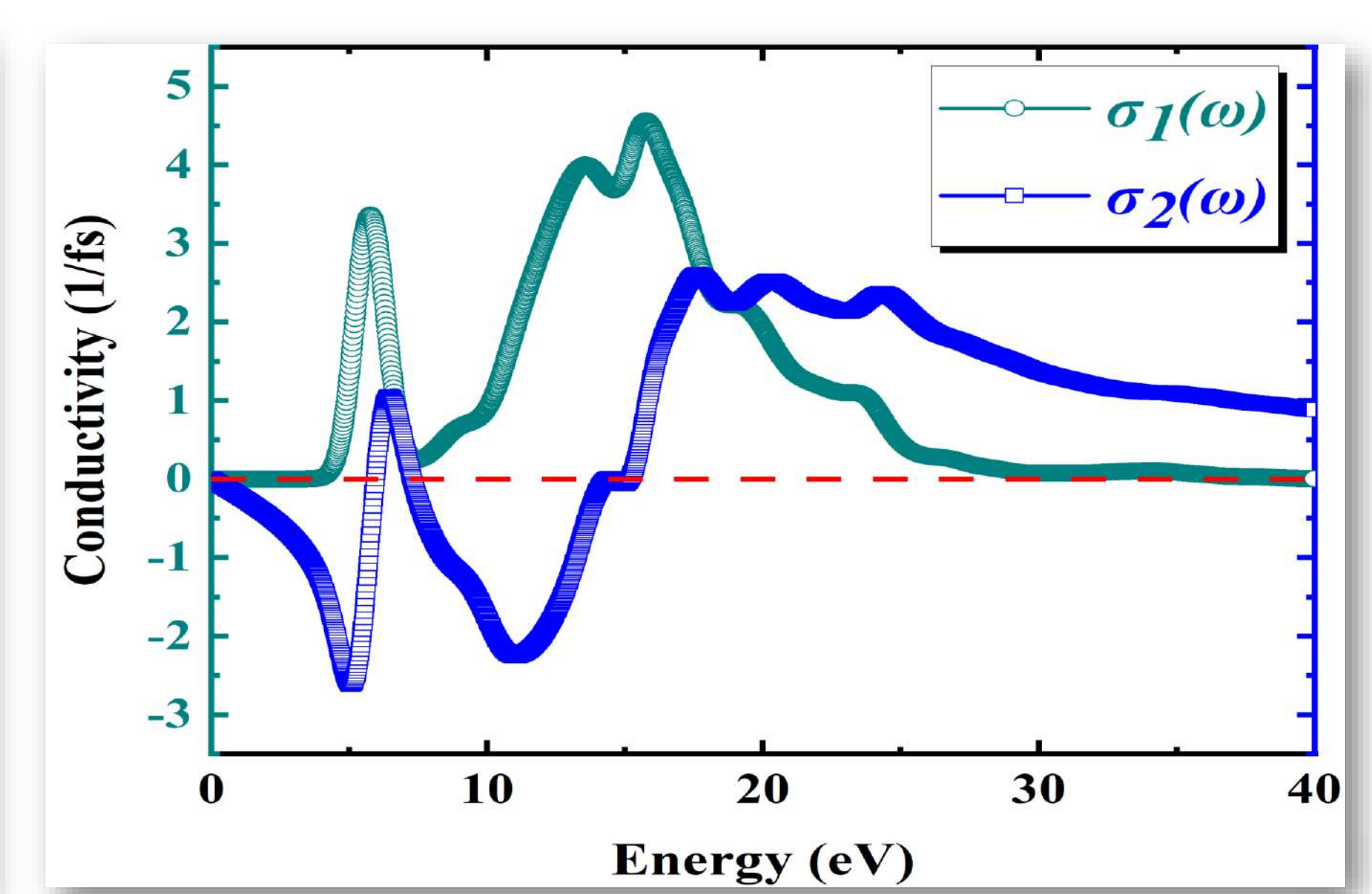


Figure 4. Conductivity

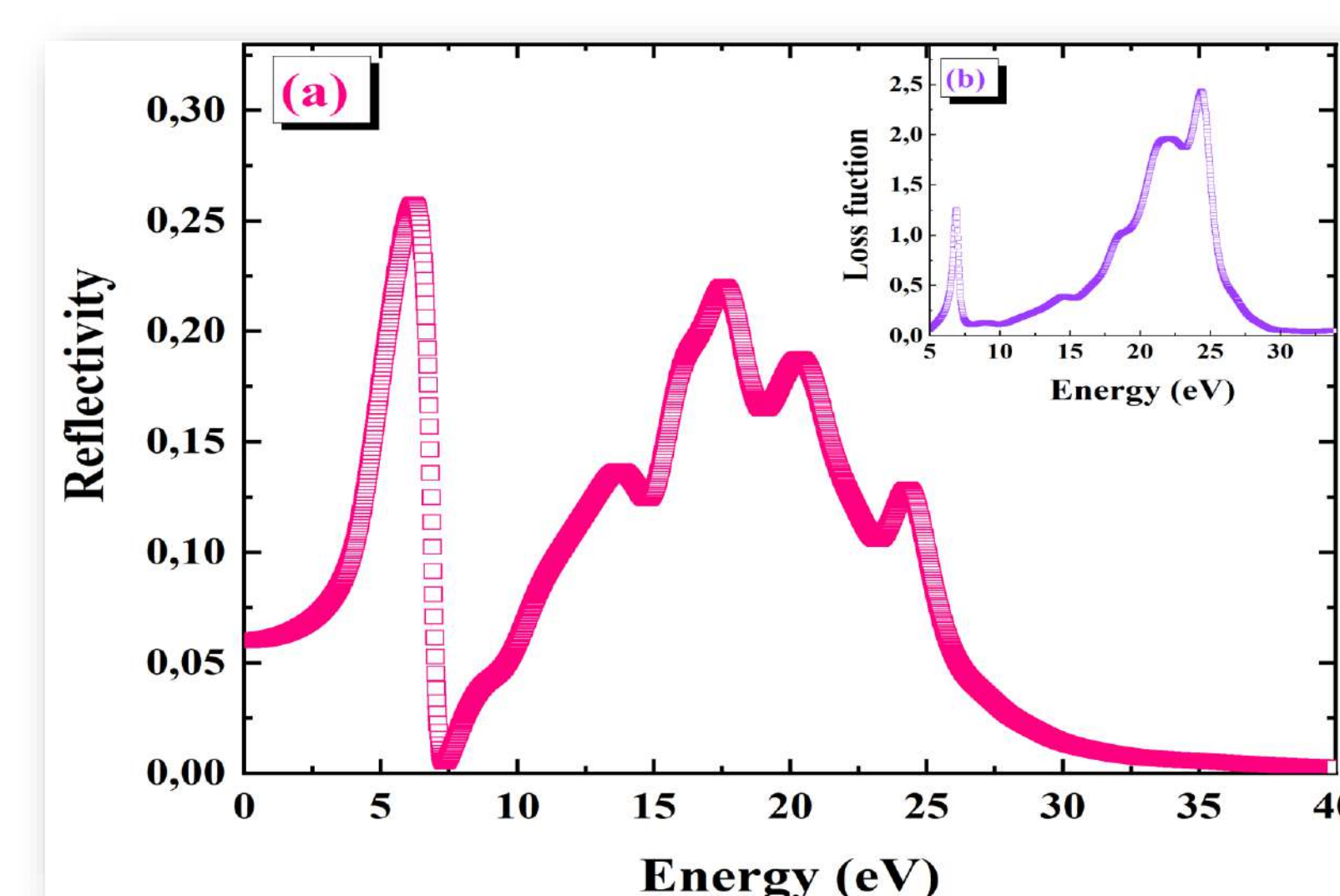


Figure 5. Reflectivity

4. CONCLUSIONS

- The (HCNO)₁₂ fragment is a reliable model for studying tropoelastin.
- It has insulating electronic properties, intense UV absorption, and anisotropic elastic behavior.
- The use of DFT in biomaterials shows promise, helping to design biomimetic materials for implants, prosthetics, and devices compatible with biological tissues.

Acknowledgments

